Two-Photon Physics at a Lepton-Proton Collider at CERN

Hamzeh Khanpour
Faculty of Physics and Applied Computer Science, AGH University of Krakow, Kraków, Poland

Contribution related to the "Ultra-cold muons for the LHC" workshop, Kraków, September 2026

Abstract

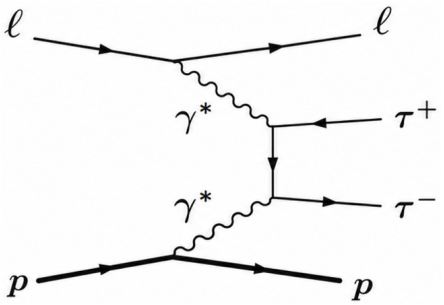
High-energy photon-photon scattering is a complementary component of the physics programme at future CERN lepton-proton facilities. We examine the equivalent-photon description of elastic and proton-dissociative two-photon reactions and investigate their physics potential at the Large Hadron-electron Collider (LHeC) and the proposed muon-proton collider (LHμC). The photon-photon luminosity determines the available rates for hard two-photon subprocesses. Two representative precision benchmarks are emphasized: tau-lepton pair production as a probe of anomalous electromagnetic dipole interactions, and W-boson pair production as a test of the electroweak quartic gauge sector. We examine how the higher collision energies and harder photon-photon luminosity spectra of the LHμC, compared with the LHeC, extend the accessible high-invariant-mass region and can enhance sensitivity to energy-dependent beyond Standard Model contributions. Loop-induced neutral gauge-boson production, top-quark pairs, charged electroweak states, and light-by-light scattering provide additional physics opportunities. Quantitative sensitivity projections require consistent photon-emission selections, effective-field-theory validity checks, and realistic detector-level studies.

1 Introduction

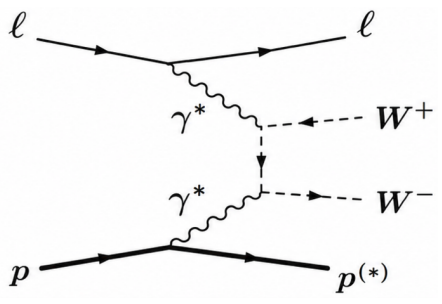
Photon-photon fusion offers a particularly transparent route to the production of electrically charged and electroweakly coupled final states. A relativistic lepton and proton radiate photons, which subsequently undergo a hard collision, $\gamma\gamma \to X$. The radiating particles may remain intact, giving a fully exclusive topology, or the proton may dissociate into a low- or high-mass hadronic system. This separation of electromagnetic emission and the short-distance subprocess underlies the equivalent photon approximation (EPA) [1, 2].

The proposed Large Hadron-electron Collider (LHeC) combines a lepton beam of order 50 GeV with the 7 TeV LHC proton beam, yielding $\sqrt{s_{ep}} \simeq 1.2$ TeV [3, 4]. An electron-proton environment is especially attractive for exclusive studies because it has one rather than two colored incoming beams and is expected to have much less event pileup than high-luminosity proton-proton collisions. Neither advantage removes detector inefficiencies, proton dissociation or all soft-interaction uncertainties, but both simplify the experimental interpretation of photon-fusion signatures [5, 6]. A complementary LHμC concept employs energetic positive-muon beams colliding with the LHC proton beam. The recent proposal of Ref. [7] emphasizes muon-beam energies of 1 and 2 TeV, corresponding to centre-of-mass energies of $\sqrt{s_{\mu p}} \simeq 5.3$ and 7.5 TeV, respectively, with a projected average luminosity approaching $10^{34}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$.

In this work, we consider three LHμC configurations at $\sqrt{s_{\mu p}} \simeq 3.7$, 5.3, and 7.5 TeV and investigate their two-photon physics potential in comparison with the LHeC. We examine the equivalent photon approximation (EPA) and the energy dependence of the photon-photon luminosity spectrum, emphasizing the enhanced high-invariant-mass reach of the LHμC and its implications for precision electroweak measurements and searches for physics beyond the Standard Model (BSM). Particular attention is devoted to two representative processes: $\gamma\gamma \to \tau^+\tau^-$, which provides sensitivity to anomalous



(a) $\gamma\gamma \to \tau^+\tau^-$



(b) $\gamma\gamma \to W^+W^-$

Figure 1: Representative Standard Model Feynman diagrams for photon-induced $\tau^+\tau^-$ (left) and W^+W^- (right) production in lepton–proton collisions, $\ell p \to \ell X p^{(*)}$, with $\ell = e, \mu^+$.

tau electromagnetic dipole moments, and $\gamma\gamma \to W^+W^-$, which probes the electroweak quartic gauge sector. Representative Standard Model (SM) Feynman diagrams for these photon-induced processes in lepton-proton collisions are shown in Fig. 1. Additional channels, including neutral gauge-boson pairs, top-quark pairs, and charged BSM states, further extend the physics opportunities, particularly in the high-energy region where the harder photon-photon luminosity spectrum of the LHμC offers significant advantages over the LHeC.

2 Theoretical framework for two-photon interactions

2.1 Equivalent photon approximation

For $\ell = e^-$ or μ^+, the generic reaction is $\ell p \to \ell + X + p^{(*)}$. In the quasi-real limit the cross section can be written as a convolution of photon spectra and the elementary photon-photon cross section,

$$\sigma_{\ell p \to \ell X p^{(*)}} = \int_0^1 dy_\ell \int_0^1 dy_p \, \Phi_\ell(y_\ell)\Phi_p(y_p) \, \hat{\sigma}_{\gamma\gamma\to X}(W_{\gamma\gamma}), \tag{1}$$

where y_i denotes the fraction of the incoming particle's longitudinal momentum carried by the emitted photon, and $\Phi_i(y_i) = dN_{\gamma/i}/dy_i$ represents the equivalent-photon flux of the emitting particle $i = \ell, p$, with $\ell = e, \mu$. Thus, $\Phi_\ell(y_\ell)$ and $\Phi_p(y_p)$ describe the photon spectra associated with the lepton and proton beams, respectively. The quantity $\hat{\sigma}_{\gamma\gamma\to X}(W_{\gamma\gamma})$ denotes the elementary cross section for the hard photon-photon subprocess $\gamma\gamma \to X$, evaluated at $W_{\gamma\gamma}$. Here $W_{\gamma\gamma}$ denotes the diphoton invariant mass, with $W^2_{\gamma\gamma} \simeq y_\ell y_p s_{\ell p}$ for collinear, quasi-real emission. Its natural generalization is

$$\begin{aligned} \sigma_{\ell p} &= \int_{W_{\rm thr}}^{\sqrt{s_{\ell p}}} dW \, S_{\gamma\gamma}(W) \, \hat{\sigma}_{\gamma\gamma\to X}(W), \\ S_{\gamma\gamma}(W) &= \frac{2W}{s_{\ell p}} \int_{W^2/s_{\ell p}}^1 \frac{dy_\ell}{y_\ell} \, \Phi_\ell(y_\ell)\Phi_p\left(\frac{W^2}{s_{\ell p} y_\ell}\right). \end{aligned} \tag{2}$$

Thus, $S_{\gamma\gamma}(W)$ has dimensions of inverse energy, and its integral over W represents the dimensionless effective photon-photon luminosity. This expression is derived within the collinear equivalent-photon approximation, where $W^2_{\gamma\gamma} \simeq y_\ell y_p s_{\ell p}$. For non-negligible photon virtualities, corrections to the two-photon invariant-mass kinematics and the virtuality dependence of the hard subprocess must be taken into account, as discussed in Sec.2.3.

2.2 Equivalent-photon spectra and photon virtuality

For a point-like lepton of mass m_ℓ, an illustrative virtuality-integrated EPA spectrum is [1]

$$\begin{aligned}\Phi_\ell(y) &= \frac{\alpha}{2\pi}\left[\frac{1+(1-y)^2}{y}\ln\frac{Q^2_{\max}}{Q^2_{\min}} - \frac{2(1-y)}{y}\left(1-\frac{Q^2_{\min}}{Q^2_{\max}}\right)\right], \\ Q^2_{\min} &\simeq \frac{m_\ell^2 y^2}{1-y}.\end{aligned} \tag{3}$$

for $Q^2_{\min} < Q^2_{\max}$, with a vanishing flux otherwise. The muon mass increases $Q^2_{\min}$ relative to an electron at fixed y: the gain from a muon collider therefore comes from its accessible beam energy and usable phase space, not from a universally larger flux at fixed energy. For an elastic proton, an often used representation is

$$\Phi_p^{\rm el}(y) = \frac{\alpha}{\pi y}\int_{Q^2_{\min}}^{Q^2_{\max}}\frac{dQ^2}{Q^2}\left[(1-y)\left(1-\frac{Q^2_{\min}}{Q^2}\right)F_E(Q^2) + \frac{y^2}{2}F_M(Q^2)\right], \tag{4}$$

where $F_E = (4m_p^2 G_E^2 + Q^2 G_M^2)/(4m_p^2 + Q^2)$ and $F_M = G_M^2$ involve the Sachs electromagnetic form factors [2]. Here $Q^2 = -q^2 > 0$ denotes the photon virtuality, and the integration bounds must be matched to the tagging or acceptance cuts. For the proton-dissociative channel $p \to \gamma^* + N^*$, the photon spectrum can instead be expressed through measured inclusive electromagnetic structure functions $F_2(x, Q^2)$ and $F_L(x, Q^2)$ [8, 9]. With M_N the dissociation mass, $x = Q_p^2/(M_N^2 - m_p^2 + Q_p^2)$, so that

$$M_N^2 - m_p^2 = Q_p^2\left(\frac{1}{x} - 1\right). \tag{5}$$

Extending the allowed proton-dissociation mass M_N and photon virtuality Q_p^2 ranges can increase the photon-induced production cross section by enlarging the accessible inelastic phase space. However, the resulting final state contains a hadronic dissociation system, and the corresponding photon flux depends on the proton electromagnetic structure functions and the imposed kinematic selections. Consequently, elastic and proton-dissociative contributions must be treated consistently and distinguished when comparing the LHeC and LHμC predictions. In particular, an inclusive photon PDF, which may contain both elastic and inelastic contributions, cannot be directly substituted for a proton-tagged elastic photon flux without accounting for the different final-state definitions and acceptance requirements.

2.3 Validity of the approximation

For $Q_\ell^2, Q_p^2 \ll W_{\gamma\gamma}^2$, transverse quasi-real photons dominate, and Eqs. (1)-(2) provide a well-motivated factorized description of photon-induced lepton-proton interactions. The commonly used high-energy relation $W_{\gamma\gamma}^2 \simeq y_\ell y_p s_{\ell p}$, however, neglects kinematic corrections associated with photon virtualities and proton dissociation. Within the kinematic conventions adopted in the present EPA treatment, the corrected expressions for the two-photon invariant mass are

$$\begin{aligned}W^2_{\gamma\gamma,\rm el} &= -Q_\ell^2 + y_\ell y_p s_{\ell p}, \\ W^2_{\gamma\gamma,\rm inel} &= -Q_\ell^2 - Q_p^2 + y_\ell y_p s_{\ell p} \\ &\quad + \frac{Q_\ell^2}{s_{\ell p}}\left(Q_p^2 + M_N^2 - m_p^2\right).\end{aligned} \tag{6}$$

Table 1: Representative LHeC and LHμC benchmark configurations, assuming a 7 TeV proton beam [4, 7, 10].

Configuration	E_ℓ [GeV]	$\sqrt{s_{\ell p}}$ [TeV]
LHeC ($e^- p$)	20	0.75
LHeC ($e^- p$)	50	1.18
LHμC (option 1)	500	3.74
LHμC (baseline)	1000	5.29
LHμC (option 2)	2000	7.48

Here Q_ℓ^2 and Q_p^2 denote the virtualities of the photons emitted by the lepton and proton, respectively, y_ℓ and y_p are their longitudinal energy fractions, and $s_{\ell p}$ is the squared lepton-proton centre-of-mass energy. The mass M_N characterizes the proton-dissociative hadronic system, while m_p denotes the proton mass. The elastic expression explicitly retains the lepton-side virtuality correction, whereas the proton-dissociative expression also depends on Q_p^2 and M_N.

Larger photon virtualities can modify the $\gamma^*\gamma^*$ matrix elements, accessible phase space, and contributions from longitudinal photon polarizations. Consequently, full virtuality-dependent calculations become preferable when $Q_{\ell,p}^2$ is not negligible compared with $W_{\gamma\gamma}^2$. Comparisons with matrix-element generators can yield percent-level agreement for elastic production when consistent kinematic definitions and phase-space selections are employed, while proton-dissociative predictions are more sensitive to the allowed M_N and Q_p^2 ranges. Accurate reconstruction of $W_{\gamma\gamma}$ and explicit control of the photon-virtuality assumptions are therefore essential for reliable comparisons between the LHeC and LHμC configurations.

3 Photon-photon luminosity at the LHeC and LHμC

Table 1 summarizes the representative beam configurations considered in this work for the LHeC and the proposed LHμC [4, 10, 7]. Assuming a 7 TeV proton beam, the corresponding centre-of-mass energies are obtained from the high-energy approximation $s_{\ell p} \simeq 4E_\ell E_p$. For the LH$\mu$C, we consider three muon-beam energies: 500 GeV (option 1), 1 TeV (baseline), and 2 TeV (option 2), corresponding to $\sqrt{s_{\mu p}} \simeq 3.7$, 5.3, and 7.5 TeV, respectively. The baseline and higher-energy configurations are emphasized in the recent machine proposal of Ref. [7]. Increasing the muon-beam energy extends the accessible photon-photon invariant-mass range, while the shape and normalization of the photon-photon luminosity spectrum also depend on the photon fluxes and the imposed kinematic selections.

As illustrated in Fig. 2, the elastic photon-photon luminosity spectrum decreases rapidly with increasing invariant mass W. However, the higher lepton-beam energies available at the LHμC result in a harder spectrum than at the LHeC. Already for $E_\mu = 500$ GeV, the luminosity is enhanced in the high-W region, while the 1 TeV muon option extends this advantage further. This enhancement is particularly relevant for the production of heavy final states and for probing energy-dependent contributions from anomalous interactions. In the region of larger W, the increase in the kinematic energy scale compensates for the steeper suppression associated with high photon momentum fractions. The quantitative ratio is nevertheless dependent on the allowed Q^2 range, proton emission mode and W selection, and the plotted example must not be used as an experimentally validated luminosity forecast.

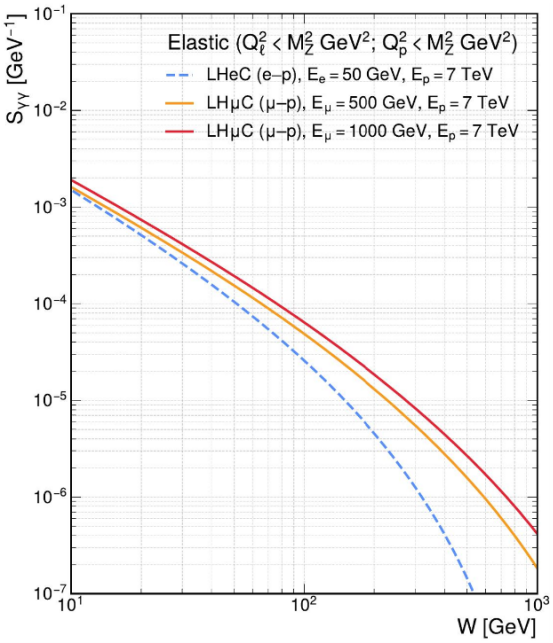


Figure 2: Elastic photon-photon luminosity spectrum $S_{\gamma\gamma}(W)$ as a function of the two-photon invariant mass W at the LHeC ($E_e = 50$ GeV) and LHμC ($E_\mu = 500$ and 1000 GeV), assuming a 7 TeV proton beam. The higher-energy muon-proton configurations provide an enhanced photon-photon luminosity in the large-W region, extending the kinematic reach for heavy final states.

Table 2: Elastic photon-induced production cross sections for $\tau^+\tau^-$ and W^+W^- pairs at the LHeC and LHμC benchmark configurations. The quoted values correspond to integrated $\ell p \to \ell X p$ production cross sections, with $X = \tau^+\tau^-$ or W^+W^-, before decay branching fractions and detector effects.

Collider	$\sqrt{s_{\ell p}}$ [TeV]	$\sigma^{\rm el}_{\tau\tau}$ [pb]	$\sigma^{\rm el}_{WW}$ [pb]
LHeC	1.2	50.40	0.037
LHμC	3.7	58.5	0.166
LHμC	5.3	70.3	0.258

4 Tau-pair production and electromagnetic moments

4.1 Standard-Model production

The tree-level process $\gamma\gamma \to \tau^+\tau^-$ is governed by QED t- and u-channel tau exchange. The relatively low production threshold makes this channel valuable for photon-luminosity calibration, while the large accessible invariant mass enables a test of the $\gamma\tau\tau$ vertex well above the tau mass. The first observation of photon-induced tau-pair production in pp collisions, together with bounds on tau electromagnetic moments, has already demonstrated the utility of this mechanism [11].

The elastic photon-induced cross sections presented in Table 2 illustrate the energy dependence of tau-pair production at the LHeC and LHμC. The total cross section increases from 50.40 pb at the LHeC ($\sqrt{s_{ep}} \simeq 1.2$ TeV) to 58.5 pb and 70.3 pb at the LHμC configurations with $\sqrt{s_{\mu p}} \simeq 3.7$ and 5.3 TeV, respectively. Although the increase in the integrated cross section is relatively modest, the higher collision energies extend the accessible diphoton invariant-mass range. The photon-photon luminosity spectra shown in Fig. 2 further illustrate the enhanced high-invariant-mass reach of the LHμC compared with the LHeC. This feature is particularly relevant for the high-mass tail

of the $M_{\tau\tau}$ distribution, where contributions from anomalous tau electromagnetic interactions may become more pronounced.

4.2 Dipole interactions and their EFT interpretation

The coupling of an on-shell tau lepton to a quasi-real photon can be modified by anomalous electromagnetic dipole interactions,

$$\mathcal{L}^{\rm dip}_{\tau\tau\gamma} = \frac{e\,\delta a_\tau}{4m_\tau}\bar\tau\sigma^{\mu\nu}\tau F_{\mu\nu} - \frac{i\,d_\tau}{2}\bar\tau\sigma^{\mu\nu}\gamma_5\tau F_{\mu\nu}, \tag{7}$$

where $\delta a_\tau = a_\tau - a_\tau^{\rm SM}$ denotes the new-physics contribution to the anomalous magnetic moment, $a_\tau = (g_\tau - 2)/2$, and d_τ is the electric dipole moment. The magnetic interaction is CP even, whereas the electric dipole term is CP odd. In the Standard Model effective field theory (SMEFT), these interactions arise from the electroweak-gauge-invariant dimension-six operators [11, 12]

$$\begin{aligned}\mathcal{O}^{33}_{eB} &= (\bar L_3\sigma^{\mu\nu}\tau_R)HB_{\mu\nu},\\ \mathcal{O}^{33}_{eW} &= (\bar L_3\sigma^{\mu\nu}\tau_R)\sigma^I HW^I_{\mu\nu},\end{aligned} \tag{8}$$

with $\Delta\mathcal{L} = (C^{33}_{eB}\mathcal{O}^{33}_{eB} + C^{33}_{eW}\mathcal{O}^{33}_{eW})/\Lambda^2 + \text{h.c.}$. Here L_3 is the third-generation left-handed lepton doublet, H is the Higgs doublet, and $B_{\mu\nu}$ and $W^I_{\mu\nu}$ are the electroweak field strengths; σ^I are the Pauli matrices. After electroweak symmetry breaking, the photon-dipole combination is

$$\begin{aligned}C_{\tau\gamma} &= c_W C^{33}_{eB} - s_W C^{33}_{eW},\\ \delta a_\tau &= \frac{2\sqrt{2}\,m_\tau v}{e\Lambda^2}\,\mathrm{Re}\,C_{\tau\gamma},\\ d_\tau &= -\frac{\sqrt{2}\,v}{\Lambda^2}\,\mathrm{Im}\,C_{\tau\gamma},\end{aligned} \tag{9}$$

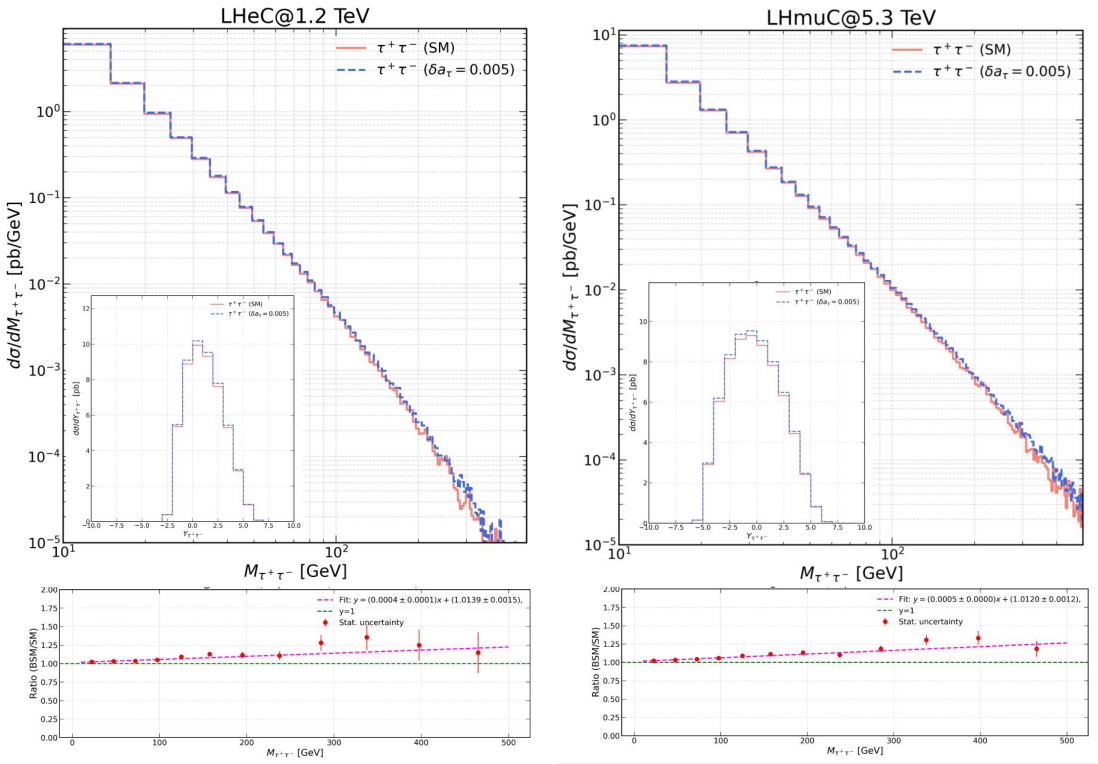
where $s_W = \sin\theta_W$, $c_W = \cos\theta_W$ and $v \simeq 246\,\text{GeV}$. The sign in the final line follows the electric-dipole convention of Eq. (7).

4.3 High-mass distributions and sensitivity

At leading order in the anomalous couplings, the $\gamma\gamma \to \tau^+\tau^-$ amplitude may be expressed schematically as $\mathcal{M} \simeq \mathcal{M}_{\rm SM} + \delta a_\tau\mathcal{M}_a + d_\tau\mathcal{M}_d$. The resulting differential rate contains interference with the SM amplitude and terms quadratic in the dipole couplings; additional higher powers can arise when both photon–tau vertices contain anomalous insertions. The magnetic dipole interaction can modify CP-even invariant-mass distributions through interference, whereas the interference linear in d_τ cancels for CP-even observables with CP-symmetric phase-space integration. Dedicated CP-odd observables can instead provide linear sensitivity to the electric dipole moment.

Figure 3 compares the Standard Model prediction with the illustrative deformation $\delta a_\tau = 0.005$ in the $\tau^+\tau^-$ invariant-mass spectrum at the LHeC ($\sqrt{s_{ep}} \simeq 1.2\,\text{TeV}$) and the LH$\mu$C baseline ($\sqrt{s_{\mu p}} \simeq 5.3\,\text{TeV}$). The spectra are dominated by the lower-mass region, where the SM and anomalous-coupling curves nearly coincide. The ratio panels show that the relative deformation grows towards larger $M_{\tau\tau}$, where the dipole interaction becomes more influential. The higher-energy LHμC configuration accesses a harder photon–photon spectrum and offers a potentially more effective high-mass lever arm than the LHeC, subject to luminosity and acceptance.

For context, CMS photon-induced τ-pair production and ATLAS high-mass τ-pair measurements constrain the *total* anomalous magnetic moment to $-0.0022 < a_\tau < 0.0041$



(a) LHeC, $\sqrt{s_{ep}} \simeq 1.2\,\text{TeV}$ (b) LHμC, $\sqrt{s_{\mu p}} \simeq 5.3\,\text{TeV}$

Figure 3: Photon-induced $\tau^+\tau^-$ invariant-mass spectra for the Standard Model and the illustrative anomalous magnetic dipole benchmark $\delta a_\tau = 0.005$. The lower panels display the ratio of the anomalous-coupling to the SM prediction, highlighting the increasing relative impact at large $M_{\tau\tau}$. The insets show the τ-pair rapidity distributions. These are benchmark-level differential predictions, not detector-level sensitivity limits.

and $-0.0024 < a_\tau < 0.0047$, respectively, at 95% confidence level [11, 13]. The choice $\delta a_\tau = 0.005$ in Fig. 3 is therefore intended to illustrate the spectral response, not to represent a new exclusion limit or a presently allowed one-coefficient benchmark. Quantitative projections for the LHμC require realistic integrated luminosities, tau-decay reconstruction, background and systematic-uncertainty modelling, and checks of EFT validity. Extending the same differential analysis to the 3.7 and 7.5 TeV LHμC configurations will allow a consistent comparison of all three scenarios with the LHeC.

5 W-boson pairs and anomalous quartic gauge couplings

5.1 Standard-Model $\gamma\gamma \to W^+W^-$

At leading order $\gamma\gamma \to W^+W^-$ proceeds through charged-W exchange in the t and u channels, together with the four-point $\gamma\gamma W^+W^-$ vertex. Gauge invariance ties these amplitudes together. Unlike massive charged scalar or fermion-pair production, the unpolarized high-energy vector-pair subprocess retains a sizable rate, making it an unusually productive probe of the electroweak gauge sector [6]. Its threshold is $W_{\text{thr}} = 2m_W$, substantially higher than for tau pairs.

For the elastic benchmarks summarized in Table 2, the W^+W^- production cross section at the LHμC exceeds the corresponding LHeC reference value by factors of approximately 4.5 and 7.0 at $\sqrt{s} = 3.7$ and 5.3 TeV, respectively. Assuming an illustrative integrated luminosity of $250\,\text{fb}^{-1}$ for each configuration, the corresponding numbers of

produced W-boson pairs are 9.25×10^3 at the LHeC, compared with 4.15×10^4 and 6.45×10^4 at the LHμC for $\sqrt{s} = 3.7$ and 5.3 TeV, respectively, before accounting for branching fractions, detector acceptance, and reconstruction efficiencies.

These results illustrate the substantial enhancement of photon-induced electroweak production at the LHμC, driven by its higher collision energy and harder photon-photon luminosity spectrum. The increased event rates, together with the extended high-invariant-mass reach, provide promising opportunities for more precise measurements and improved sensitivity to anomalous electroweak interactions. Quantitative sensitivity projections, however, require consistent event selections, background estimates, and detector-level reconstruction.

5.2 Dimension-eight quartic interactions

Anomalous quartic gauge couplings (aQGCs) are conveniently parameterized in a linearly realized electroweak EFT using higher-dimensional operators. A generic representation is

$$\mathcal{L}_{\mathrm{EFT}} = \mathcal{L}_{\mathrm{SM}} + \sum_i \frac{f_i}{\Lambda^4}\mathcal{O}_i^{(8)} + \cdots, \qquad \sigma = \sigma_{\mathrm{SM}} + \sum_i \frac{f_i}{\Lambda^4}\sigma_{\mathrm{int},i} + \sum_{ij} \frac{f_i f_j}{\Lambda^8}\sigma_{\mathrm{quad},ij}. \quad (10)$$

The commonly used S, M and T operator families organize contributions involving covariant Higgs derivatives and electroweak field strengths [14, 15]. For example, in one standard convention $\mathcal{O}_{M2} = (B_{\mu\nu}B^{\mu\nu})[(D_\beta H)^\dagger D^\beta H]$. A benchmark such as $f_{M2}/\Lambda^4 = 1\,\mathrm{TeV}^{-4}$ changes the amplitude, but is *not* by itself an experimental constraint. Equation (10) also makes clear that large positive high-mass deviations can be dominated by the quadratic EFT term, potentially in a regime where the truncation becomes unreliable.

5.3 Comparison and EFT domain of validity

The principal sensitivity of $\gamma\gamma \to WW$ to anomalous contributions occurs in high-M_{WW} bins, where increased hard-scattering energy may offset declining photon luminosity. A collider-to-collider comparison must hold the EFT coefficients, final-state acceptance, photon virtuality cuts and proton-emission topology fixed. In particular, the original LHμC aQGC distribution used a 5.3 TeV collider configuration, while the elastic SM cross-section number just quoted corresponds to 3.7 TeV; one must not mix them into a single numerical limit. Furthermore, an LHC proton–proton comparison is affected by proton survival probabilities, pileup and tagging conditions that have no simple counterpart in ℓp scattering.

An experimentally interpretable analysis would report the selected reconstructed final state, detector response, background template, correlated uncertainties and treatment of parameter degeneracies. It would also specify an energy domain such that the operator expansion remains controlled. A visible ratio $[d\sigma/dM_{WW}]_{\mathrm{EFT}}/[d\sigma/dM_{WW}]_{\mathrm{SM}}$ signals a potentially discriminating observable; it is not equivalent to a detector-level 95% confidence interval. This distinction is central when comparing upgraded LHμC energies with earlier baseline studies.

6 Additional Standard-Model and BSM opportunities

Beyond tau and W pairs, several photon-fusion processes become particularly interesting as the accessible diphoton invariant mass increases. The neutral final state $\gamma\gamma \to ZZ$ is

loop-induced in the SM, making its high-mass spectrum a promising probe of anomalous neutral quartic gauge interactions. The process $\gamma\gamma \to t\bar{t}$ provides a complementary electromagnetic production mechanism for the heaviest SM fermion and offers sensitivity to possible deviations in its photon couplings. With a production threshold near $2m_t$, its rate depends strongly on the high-energy tail of $S_{\gamma\gamma}$ and must be distinguished from the substantially larger γg top-quark photoproduction channel.

Charged electroweak states provide further opportunities for probing physics beyond the SM. Higgsino-like chargino and charged-slepton pairs can be produced through $\gamma\gamma \to \widetilde{H}^+\widetilde{H}^-$ and $\gamma\gamma \to \widetilde{\ell}^+\widetilde{\ell}^-$, respectively. Their leading electromagnetic interactions are determined by the electric charge, while their spin and mass govern the threshold behaviour and angular distributions. Compressed mass spectra remain experimentally challenging because the visible decay products may be soft and the missing-momentum signatures require accurate event reconstruction. Existing LHC searches already constrain parts of this parameter space, making photon-induced production at lepton–hadron colliders a complementary approach.

Finally, the loop-induced processes $\gamma\gamma \to \gamma\gamma$ and $\gamma\gamma \to H$ provide additional probes of charged virtual particles and potential BSM contributions. Their relatively small production rates emphasize the importance of controlling photon spectra, exclusive-event selections, and backgrounds. Compared with the LHeC, the higher centre-of-mass energies available at the LHμC lead to a harder photon–photon luminosity spectrum, enhancing the accessible high-W region. This advantage is particularly relevant for heavy final states, where production thresholds strongly suppress the LHeC rates, and for anomalous interactions whose effects grow with energy. Consequently, the LHμC offers the potential for substantially improved mass reach and enhanced sensitivity to electroweak and BSM interactions, subject to realistic luminosity, acceptance, and background assumptions. These additional channels significantly broaden the photon-induced physics programme while complementing the precision studies of tau and W-boson pairs.

7 Experimental considerations and future prospects

A reliable photon-fusion analysis must simultaneously account for the outgoing-lepton acceptance, central-system reconstruction, and proton final state. Elastic channels may exploit forward proton tagging, large rapidity gaps, small transverse-momentum imbalance, and acoplanarity, whereas proton-dissociative channels require either explicit hadronic reconstruction or a carefully modelled inclusive treatment. Tau decays introduce neutrinos, while W-boson reconstruction involves branching fractions, combinatorial ambiguities, and missing-energy effects. Searches for new charged particles may additionally be limited by detector acceptance and reconstruction efficiencies. These experimental effects are not fully captured by generator-level EPA calculations.

The machine-detector interface for the LHμC remains under development [7]. Quantitative sensitivity studies must therefore account for muon-beam-induced backgrounds, beam optics, forward-tagging acceptance, and realistic integrated luminosities. Future investigations should combine validated photon fluxes with full matrix-element calculations where appropriate, detector simulations, and statistical analyses consistent with EFT validity requirements. The extended calculations at $\sqrt{s} = 3.7$, 5.3, and 7.5 TeV enable a systematic comparison of the LHμC physics potential across different beam-energy configurations. In particular, the enhanced high-invariant-mass photon-photon luminosity provides greater kinematic reach for heavy charged states and offers promising opportunities for improved sensitivity to anomalous electroweak interactions

relative to the LHeC.

8 Summary and conclusions

High-energy photon fusion offers a complementary electroweak and BSM physics programme at future CERN lepton-proton colliders. The EPA organizes the connection between beam-dependent photon fluxes and the hard $\gamma\gamma$ subprocess, while elastic and proton-dissociative emission must remain distinct. Compared with the LHeC, the higher beam energies attainable at the LHμC provide a harder $\gamma\gamma$ luminosity spectrum and substantially enhance the accessible high-invariant-mass region, offering improved sensitivity reach to anomalous electroweak interactions and heavy charged states. In particular, tau-pair spectra and W-pair production furnish promising probes of anomalous electromagnetic dipole moments and quartic gauge couplings, respectively, with potentially enhanced sensitivity at the LHμC. Consistent kinematic selections, reliable photon-emission models, detector response, and EFT validity checks remain indispensable for establishing quantitative sensitivity projections.

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